

Start

Configure S3 bucket: [here](#)

VPC

How to create VPC and subnets: [here](#)

First, create a VPC

The screenshot shows the AWS VPC dashboard for a VPC named 'vpc-0f2bedf344eeb994f / VPC-Company-Production-Network'. A green notification bar at the top states 'You successfully created vpc-0f2bedf344eeb994f / VPC-Company-Production-Network'. The left sidebar shows the 'VPC dashboard' with a filter by VPC. The main content area displays the VPC details in a grid format:

- VPC ID:** vpc-0f2bedf344eeb994f
- DNS resolution:** Enabled
- Main network ACL:** acl-0c47d982048bc6e0d
- IPv6 CIDR (Network border group):** -
- Encryption control ID:** -
- State:** Available
- Tenancy:** default
- Default VPC:** No
- Network Address Usage metrics:** Disabled
- Encryption control mode:** -
- Block Public Access:** Off
- DHCP option set:** dopt-0531e230b3c3a546f
- IPv4 CIDR:** 10.0.0.0/16
- Route 53 Resolver DNS Firewall rule groups:** -
- DNS hostnames:** Disabled
- Main route table:** rtb-0d8aef61a39b69776
- IPv6 pool:** -
- Owner ID:** 967883358429

Below the details, there are tabs for 'Resource map', 'CIDRs', 'Flow logs', 'Tags', and 'Integrations'. The 'Resource map' tab is selected, showing a 'Resource map' section with a 'Show all details' toggle.

Create public subnet & Private subnet:

The screenshot shows the AWS Subnets console. A green notification bar at the top states 'You have successfully created 1 subnet: subnet-0228462bb306b745f'. The main content area displays a list of subnets with the following columns: Name, Subnet ID, State, VPC, Block Public..., and IPv4 CIDR. The list includes several test subnets and two production subnets: 'Company-SupplyChain-Private-Subnet' and 'Company-Public-Subnet'.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-06ccbf48cd1c11f11	Available	vpc-0323b64209f368ca0 Testi...	Off	172.31.0.0/20
-	subnet-087cbf9d09a9148ef	Available	vpc-0323b64209f368ca0 Testi...	Off	172.31.64.0/20
-	subnet-01a2def073d7c6b6c	Available	vpc-0323b64209f368ca0 Testi...	Off	172.31.16.0/20
-	subnet-06e61a165a7d41235	Available	vpc-0323b64209f368ca0 Testi...	Off	172.31.80.0/20
Company-SupplyChain-Private-Subnet	subnet-0665b30531f3dd914	Available	vpc-0f2bedf344eeb994f VPC-...	Off	10.0.1.0/24
Company-Public-Subnet	subnet-0228462bb306b745f	Available	vpc-0f2bedf344eeb994f VPC-...	Off	10.0.2.0/24

Important Note: This is our goal.

The Internet gateway or IGW should be inside the public subnet. Anything that is inside the public subnet that has the IGW has a public IP that ANYONE can see and access.

Moreover, for a private subnet, a private instance, sometimes you need to download stuff from the Internet, but do not want to expose your own IP because you only have one IP, and it is private. Therefore, you need a NAT gateway.

The NAT gateway alone cannot access the Internet if placed in the private subnet; thus, the NAT gateway has to have access from the IGW routable back into NAT and back into the private instance.

Internet > IGW > NAT > Private Subnet/Instance

Analogy: Chef > Waiter > Customer

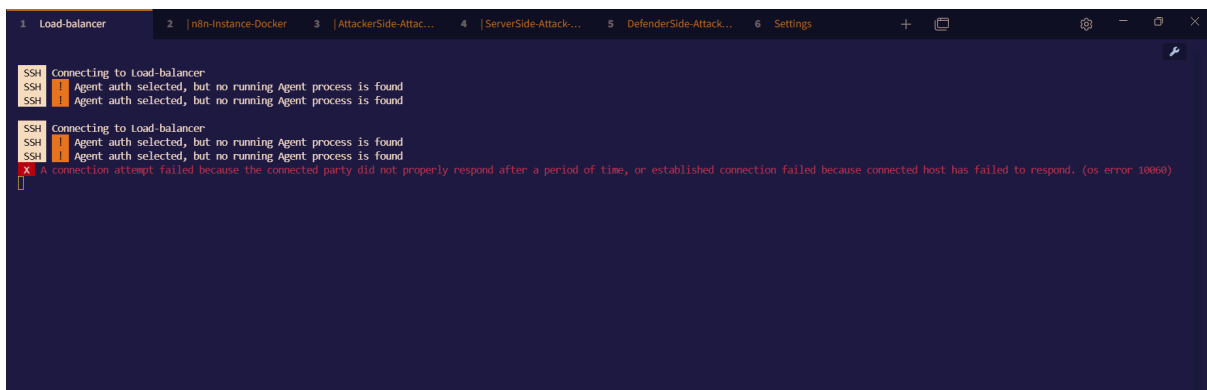
Both the public subnet & public route table have an IGW.

While both the private subnet & private route table have a NAT gateway.

In simple terms, a NAT gateway stays in the public subnet because the public subnet also has IGW or Internet access. The private subnet and its instances only take the NAT IP and its ID to use, since that is NAT's responsibility, to provide data on behalf of the private client.

Internet gateway

Initially, both public and private instances cannot connect to the Internet. So we need the Internet Gateway & Network Address Translation Gateway.



Here are the steps:

Internet gateway igw-0e9c01c66e5fc57cf successfully attached to vpc-0f2bedf344eeb994f

igw-0e9c01c66e5fc57cf / Company-Internet-Gateway

Actions

Details Info

Internet gateway ID igw-0e9c01c66e5fc57cf	State Attached	VPC ID vpc-0f2bedf344eeb994f VPC-Company-Production-Network	Owner 967883358429
--	-------------------	--	-----------------------

Tags (1) Manage tags

Search tags

Key	Value
Name	Company-Internet-Gateway

Need a route table that manages where the traffic goes:

Flow logs | **Route table** | Network ACL | CIDR reservations | Sharing | Tags

Route table: rtb-0d8aef61a39b69776

Edit route table association

Routes (1)

Filter routes

Destination	Target
10.0.0.0/16	local

From the public subnet, there's already a route table to the private network. But now we need a new routing table to connect to the Internet.

VPC > Route tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Company-Public-RouteTable

VPC
The VPC to use for this route table.
vpc-0f2bedf344eeb994f (VPC-Company-Production-Network)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
Company-Public-RouteTable

Add new tag

You can add 49 more tags.

Cancel Create route table

By default, it will connect to the private VPC IP. Now, add and connect to the Internet gateway for Internet access.

VPC > Route tables > rtb-01733b670d96944b5 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

Next, go to Public subnet and edit routable:

VPC > Subnets > subnet-0228462bb306b745f > Edit route table association

Edit route table association

Subnet route table settings

Subnet ID: subnet-0228462bb306b745f

Route table ID: rtb-01733b670d96944b5 (Company-Public-RouteTable)

Routes (2)

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-0e9c01c66e5fc57cf

[Cancel](#) [Save](#)

Now, it should work.

```
1 | n8n-Instance-Docker 2 | AttackerSide-Attac... 3 | ServerSide-Attac... 4 | DefenderSide-Attac... 5 | Settings 6 | Load-balancer + |

Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon Jan 26 07:56:51 UTC 2026

System load: 0.08      Temperature: -273.1 C
Usage of /: 25.9% of 6.71GB  Processes: 110
Memory usage: 24%      Users logged in: 0
Swap usage: 0%         IPv4 address for ens5: 10.0.2.201

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-2-201:~$
```

Now, the public subnet is public on the Internet, and the private subnet is still private.

Next, on the reverse proxy CLI (public instance), install nginx.

```
sudo apt update && sudo apt install nginx -y
```

```
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

```
sudo systemctl status nginx
```

Private Subnet

Creating a private subnet

Create an AMI image and launch it.

Make sure:

- VPC
- No public IP, since it's a private subnet

Private-Server-key

Create new key pair

▼ Network settings

VPC - required

vpc-0f2bedf344eeb994f (VPC-Company-Production-Network)

10.0.0.0/16

Subnet

subnet-0665b30531f3dd914

Company-SupplyChain-Private-Subnet

VPC: vpc-0f2bedf344eeb994f Owner: 967883358429 Availability Zone: us-east-1a (use1-az1)

Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP

Disable

Firewall (security groups)

Create security group

Select existing security group

Security group name - required

Private-Server

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . - / ! # \$ % & ' () * + , - . : ;

▼ Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0b6c6ebed2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Port and the IP range should be the same as the VPC or ONLY to the public subnet IP. In this case, it is the same as the VPC.

EC2 > Instances > Launch an instance

Private-Server

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&{}!\$*

Description - *required* [Info](#)

Private-Server

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 8080, 10.0.0.0/16)

Remove

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source type [Info](#)

Source [Info](#)

Description - *optional* [Info](#)

Custom TCP

TCP

8080

Custom

Q Add CIDR, prefix list or security group

10.0.0.0/16 X

e.g. SSH for admin desktop

Add security group rule

Advanced network configuration

Result:

As you can see, it does not have a public IP, but a private IP that can only be used within the VPC.

Instance summary for i-0bb9933e040831de1 (Private-Server) Info			Connect	Instance
Updated less than a minute ago				
Instance ID	Public IPv4 address	Private IPv4 addresses		
 i-0bb9933e040831de1	—	 10.0.2.214		
IPv6 address	Instance state	Public DNS		
—	 Running	—		
Hostname type	Private IP DNS name (IPv4 only)			
IP name: ip-10-0-2-214.ec2.internal	 ip-10-0-2-214.ec2.internal			

Only the reverse proxy instance machine that is within the same VPC can communicate with this private server or app.

Connect from the public instance to the private instance

(from Public instance)

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sg-04a5ed493f18af8b8	HTTP	TCP	80	Custom	0.0.0.0/0	Delete
-	SSH	TCP	22	Custom	58.97.200.0/16	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

The SSH IP range is your router's public IP. It is safer. Outbound: leave it as default (0.0.0.0/0)

(from private instance)

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sg-01ad5aba9d9523025	HTTP	TCP	80	Custom	sg-0e3874035fe5a7ce8	Delete
sg-0cdfa142de9bb1ab	SSH	TCP	22	Custom	sg-0e3874035fe5a7ce8	Delete

[Add rule](#)

Security Groups

- default | sg-015c62fb44416b7b7
- Load-Balancer | sg-0e3874035fe5a7ce8
- Private-Server | sg-0e6ee758106ebfc38

Prefix lists

- com.amazonaws.us-east-1.dynamodb | pl-02cd2c6b
- com.amazonaws.global.ipv6.cloudfront.origin-facing | pl-02d12e369a4312e03

Next, from your laptop, open Tabby and connect to the public instance:

```
PS C:\Users\User\Organized_Downloads> scp -i .\Load-Balancer-key.pem .\Private-Server-key.pem ubuntu@98.92.205.49:/home/ubuntu/
The authenticity of host '98.92.205.49 (98.92.205.49)' can't be established.
ED25519 key fingerprint is SHA256:R3HGaeQYFPuAALCan2gc0gkhH90Cdna7W0Ik90g20Ic.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '98.92.205.49' (ED25519) to the list of known hosts.
Private-Server-key.pem      100% 1678    5.9KB/s   00:00
PS C:\Users\User\Organized_Downloads>
```

(from Windows)

(from Tabby)

```
ubuntu@ip-10-0-1-252:~$ ls
ubuntu@ip-10-0-1-252:~$ who am i
ubuntu pts/0 2026-01-26 11:22 (58.97.219.104)
ubuntu@ip-10-0-1-252:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:f6:27:c0:d2:05 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 10.0.1.252/24 metric 100 brd 10.0.1.255 scope global dynamic ens5
        valid_lft 2148sec preferred_lft 2148sec
    inet6 fe80::f6:27ff:fec0:d205/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-10-0-1-252:~$ ls
Private-Server-key.pem
```

Next, open a terminal in a folder that contains the private-key.pem and the load-balancer.pem

Next,

```
scp -i .\load-balancer.pem private-key.pem
ubuntu@{public-IP-from-public-instance}:/home/ubuntu
```

This should upload the private-key.pem to the public instance because ONLY the public instance can communicate with the private subnet/instance because both are within the same VPC.

Inside the public instance CLI, type these commands:

```
ls
```

```
chmod 400 private-key.pem
```



```
ssh -i private-key.pem ubuntu@{private-IP-from-private-instance}
```

If you don't follow, you get these errors:

```

ubuntu@ip-10-0-1-252:~$ ls
Private-Server-key.pem
ubuntu@ip-10-0-1-252:~$ ssh ubuntu@10.0.2.37
The authenticity of host '10.0.2.37 (10.0.2.37)' can't be established.
ED25519 key fingerprint is SHA256:1M09DtC2wZnsCHge4Qczfed6S+CcqG0RrMTnSPqNCJk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.37' (ED25519) to the list of known hosts.
ubuntu@10.0.2.37: Permission denied (publickey).
ubuntu@ip-10-0-1-252:~$ ssh -i Private-Server-key.pem ubuntu@10.0.2.37
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@                WARNING: UNPROTECTED PRIVATE KEY FILE!                @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0664 for 'Private-Server-key.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "Private-Server-key.pem": bad permissions
ubuntu@10.0.2.37: Permission denied (publickey).

```

After you enter the private instance CLI, you cannot have access to the internet; thus you cannot download or update the system. That's why we need to configure NAT gateway ontop of the Internet gateway.

Without the NAT gateway, you will experience this:

```
[1] ~n8-instance-Docker | [AttackerSide-Attack...] [ServerSide-Attack...] [DefenderSide-Attack...] [Settings] [6 Load-balancer] + - -
```

```
ubuntu@ip-10-0-2-37:~$ sudo apt update & sudo apt upgrade -y  
[1] 1225  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
  
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Ign:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Ign:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease  
Ign:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Ign:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Ign:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease  
Ign:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Ign:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Ign:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease  
Err! http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Err! http://security.ubuntu.com/ubuntu noble-security InRelease  
  
Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002::1:103). - connect ((101: Network is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002::1:103). - con  
nect ((101: Network is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002::1:101). - connect ((101: Network is unreachable)) Cannot initiate the connection to security.ubunt  
.com:m:80 (2620:2d:4000::1:102). - connect ((101: Network is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2906:bcb0:0:1000::16). - connect ((101: Network is unreachable)) Cannot initia  
te e connection to security.ubuntu.com:80 (2620:2d:4002::1:101). - connect ((101: Network is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2906:bcb0:0:1000::18). - connect ((101: Networ  
k is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2906:bcb0:0:1000::17). - connect ((101: Network is unreachable)) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:  
4002::1:104) http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Err! http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease  
Err! http://security.ubuntu.com/ubuntu noble-security InRelease  
Connection timed out Could not connect to security.ubuntu.com:80 (91.189.91.83), connection timed out Could not connect to security.ubuntu.com:80 (185.125.190.81), connection timed out Could not connect to security.ubu  
ntu.com:80 (91.189.92.24), connection timed out Could not connect to security.ubuntu.com:80 (185.125.190.83), connection timed out Could not connect to security.ubuntu.com:80 (91.189.92.22), connection timed ou  
t Could not connect to security.ubuntu.com:80 (91.189.92.23), connection timed out Could not connect to security.ubuntu.com:80 (91.189.91.82), connection timed out  
Err:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c01:32a9:41b2:cfa3:904a). - connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.arch  
ive.ubuntu.com:80 (2600:f1f8:5c55:4c01:32a9:41b2:cfa3:904a). - connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c02:b55cf:883eb  
81a). - connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c00:7447:959ff:f86cd0e2). - connect ((101: Network is unreachable)) Canno  
t initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c00:2637:a2bc:ccfe:30d5). - connect ((101: Network is unreachable)) cannot initiate the connection to us-east-1.ec2.archive.ubun  
tu.com:80 (2600:f1f8:5c55:4c04:bffe:122c:b055:e884). - connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c05:17c8:ad29:43bf:e848). - c  
connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c01:eddc:fdffa:1470:afce). - connect ((101: Network is unreachable)) Cannot initiat  
he connection to us-east-1.ec2.archive.ubuntu.com:80 (2600:f1f8:5c55:4c02:7669:c120:50a2:9afa). - connect ((101: Network is unreachable)) Cannot initiate the connection to us-east-1.ec2.archive.ubuntu.com:80  
(2600:f1f8:5c55:4c03:edd9:9571f:P04:dac8). - connect ((101: Network is unreachable)) could not connect to us-east-1.ec2.archive.ubuntu.com:80 (34.201.250.36), connection timed out could not connect to us-east-1  
ec2.archive.ubuntu.com:80 (3.87.126.146), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (34.237.137.22), connection timed out could not connect to us-east-1.ec2.archive.ubunt  
u.com:80 (52.91.41.41), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (54.465.17.230), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (18.232.15  
0.247), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (54.144.148.213), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (52.207.133.243), conn  
ection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (54.87.19.168), connection timed out could not connect to us-east-1.ec2.archive.ubuntu.com:80 (3.209.10.109), connection timed out  
Err:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
```

NAT Gateway Configuration

Set up NAT Gateway: [here](#)

- Create NAT Gateway
- In the Private Subnet, & routable: add the NAT gateway IP and its ID
- Just watch the video or refer to the Important Note at the top of this documentation.

Then, restart the private instance and update again. It should work and have access to the internet.

On the private instance:

Next, `sudo apt install apache2 -y`

(Yes, the public instance uses nginx and the private uses Apache. It is normal.)

Then, configure the private instance and its html files.

```
sudo apt update
```

```
sudo apt install -y apache2
```

```
echo "<h1>Project Success: I am the Private Server</h1>" | sudo tee  
/var/www/html/index.html
```

Back to the public instance (connect the public instance to open/display publicly):

```
sudo nano /etc/nginx/sites-available/default
```

Change the original to this (or ignore the comments):

```
server {
```

```
    listen 80 default_server;
```

```
    listen [::]:80 default_server;
```

```
    root /var/www/html;
```

```
    index index.html index.htm;
```

```
server_name _;
```

1. THE REVERSE PROXY LOGIC

```
location / {
```

```
# Replace 10.0.2.XXX with your Private Instance's ACTUAL Private IP
```

```
proxy_pass http://10.0.2.XXX;
```

```
# Standard Proxy Headers
```

```
proxy_set_header Host $host;
```

```
proxy_set_header X-Real-IP $remote_addr;
```

```
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

2. THE "FAIL" LOGIC

```
# This tells Nginx: if the private server gives an error, use our custom page
```

```
proxy_intercept_errors on;
```

```
}
```

3. CUSTOM ERROR MESSAGE

```
# If the private server is down (502) or timed out (504), show this page
```

```
error_page 404 500 502 503 504 /custom_error.html;
```

```
location = /custom_error.html {
```

```
root /var/www/html;
```

```
internal;
```

```
}
```

```
}
```

Save and exit.

Create an error-handling file:

```
sudo nano /var/www/html/custom_error.html
```

Add this:

```
<html>
```

```
<body style="text-align: center; font-family: sans-serif; padding-top: 50px;">
```

```
<h1>⚠️ Something is Wrong at the Public Instance</h1>
```

```
<p>I am the Reverse Proxy. I am working, but I cannot reach the Private Web Server.  
Check /var/www/html/</p>
```

```
</body>
```

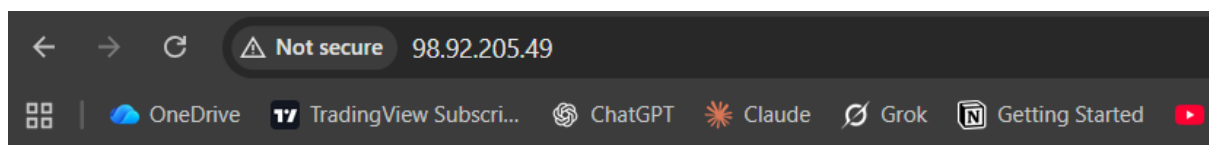
```
</html>
```

Save and exit.

Check typos: `sudo nginx -t`

Restart: `sudo systemctl restart nginx`

Lastly, get the public IP from the public instance and enter it in the search browser. You should see the text that says, “Project Success: I am the Private Server”



Project Success: I am the Private Server

Note: DO NOT OPEN THE WEBSITE USING “open address”. Instead, copy the blue-highlighted IP or copy it. Else the website won’t open or work.

i-03035f16a3cd073d6 (Load-Balancer)

Details

Status and alarms

Monitoring

Security

Networking

Storage

▼ Instance summary

Info

Instance ID

 i-03035f16a3cd073d6

Public IPv4 address

 3.239.236.34 | [open address](#) 

Verdicts:

- The IP is a public IP
 - Acts as a reverse proxy
- The text is from the private instance/subnet
 - Uses a NAT gateway on top of the Internet gateway
 - Can access the internet, but the Internet cannot access back or is unable to see

END